

MANUFACTURING ENGINEERING — DESIGN PORTFOLIO

Hydraulic Machinery & Material Handling Systems

A CAD-based design study of seven hydraulic and mechanical systems — from portable material-handling jacks to hydraulic presses and mobile-platform components.

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Seven Systems, One Design Language

Each design applies hydraulic actuation to a distinct material-handling or metal-forming task. All models are shown as isometric CAD renders with structural, hydraulic, and functional analysis.

01 **Container Lifting Jack**
Wheeled hydraulic drum/container jack

02 **Fork-Mounted Floor Jack**
Parallelogram hydraulic lift attachment

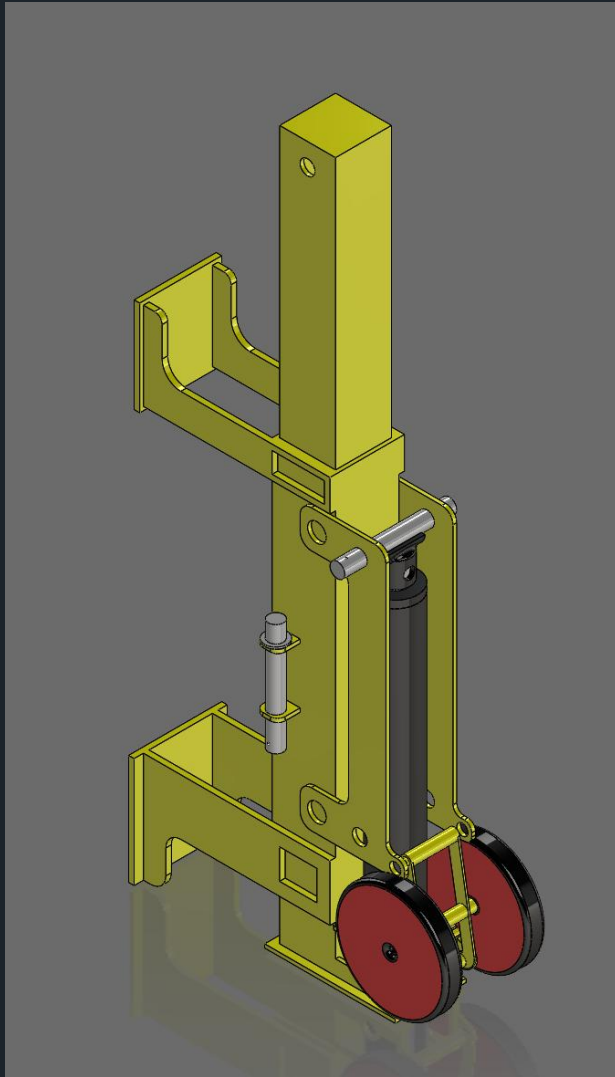
03 **Horizontal Moving Jack**
Multi-cylinder hydraulic positioning block

04 **Pipe Bending Machine**
Pedestal rotary-arm hydraulic bender

05 **Punch Press (H-Frame)**
General-purpose hydraulic shop press

06 **Sheet Metal Punch Press**
Tooled H-frame blanking/piercing press

07 **Boom-Lift Oil Tank & Mast**
Telescoping mast with reservoir base



MATERIAL HANDLING

Container / Drum Lifting Jack

A wheeled, single-ram hydraulic jack with a vertical mast and two U-shaped fork saddles — one fixed low, one sliding on the mast. The lower fork engages a drum or crate from beneath; the hydraulic ram then raises the mast-mounted upper fork so the load can be lifted clear of the floor and rolled on the twin load wheels.

KEY DESIGN FEATURES

- Dual U-fork saddles for handling loads at two different heights
- Single hydraulic ram, guided along a bolted vertical mast
- Twin large-diameter load wheels for rolling transport once lifted
- Bolted/pinned frame joints for maintenance and part replacement

800 lb
TYPICAL CAPACITY
CLASS

~130 mm
TYPICAL LIFT TRAVEL

2-wheel
ROLLING BASE

Reference specifications based on commercial 55-gal drum/container jack classes (e.g. Global Industrial, Vestil).

MATERIAL HANDLING

Fork-Mounted Floor Jack

A parallelogram lift attachment built around a central hydraulic cylinder and two boom arms. The linkage keeps the fork carriage level as it rises, while a sliding-rail base beneath the frame spreads ground contact and adds tipping stability — a configuration typical of tractor-mounted or towable pallet-fork lifters.

KEY DESIGN FEATURES

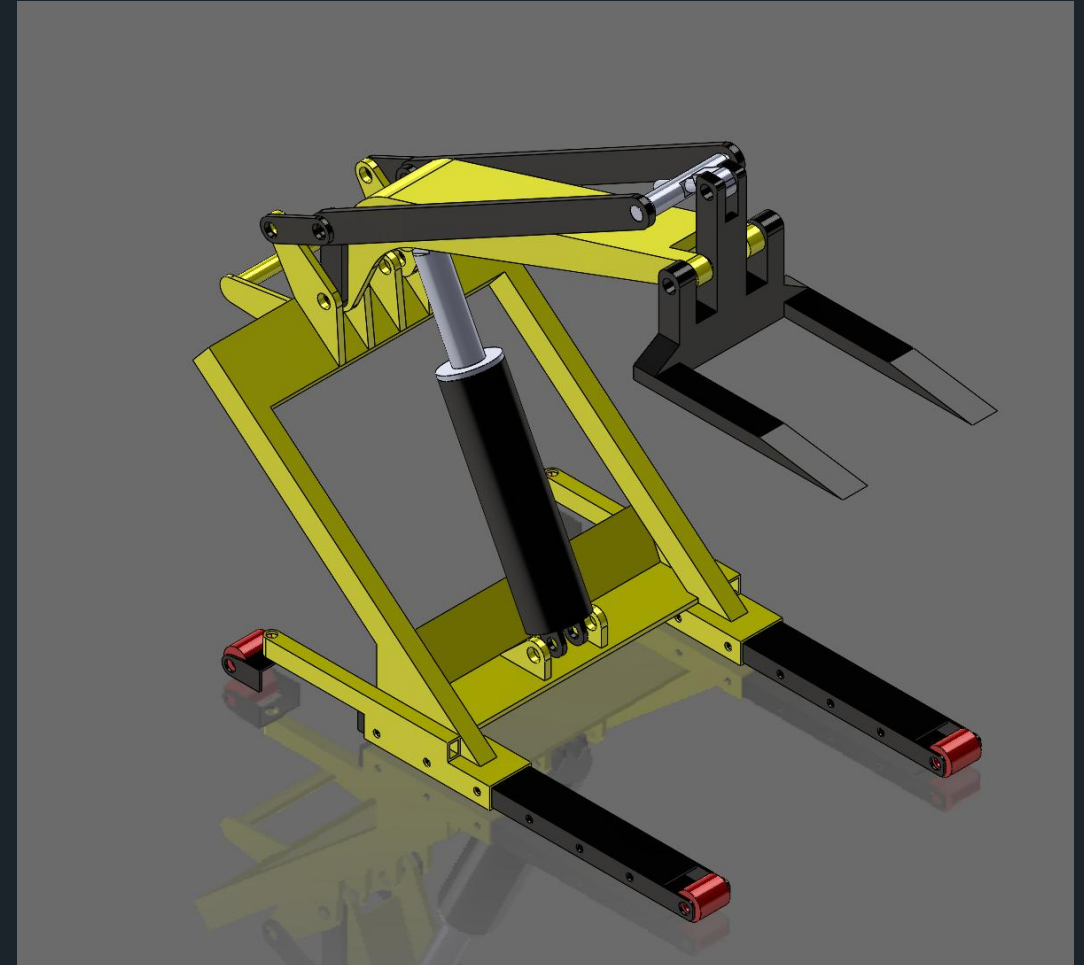
- Central hydraulic cylinder driving a parallelogram boom linkage
- Fork carriage remains level throughout the lift (no manual tilt correction)
- Sliding-rail/skid base for stability and low ground clearance
- Clevis-pinned joints at every pivot for easy field service

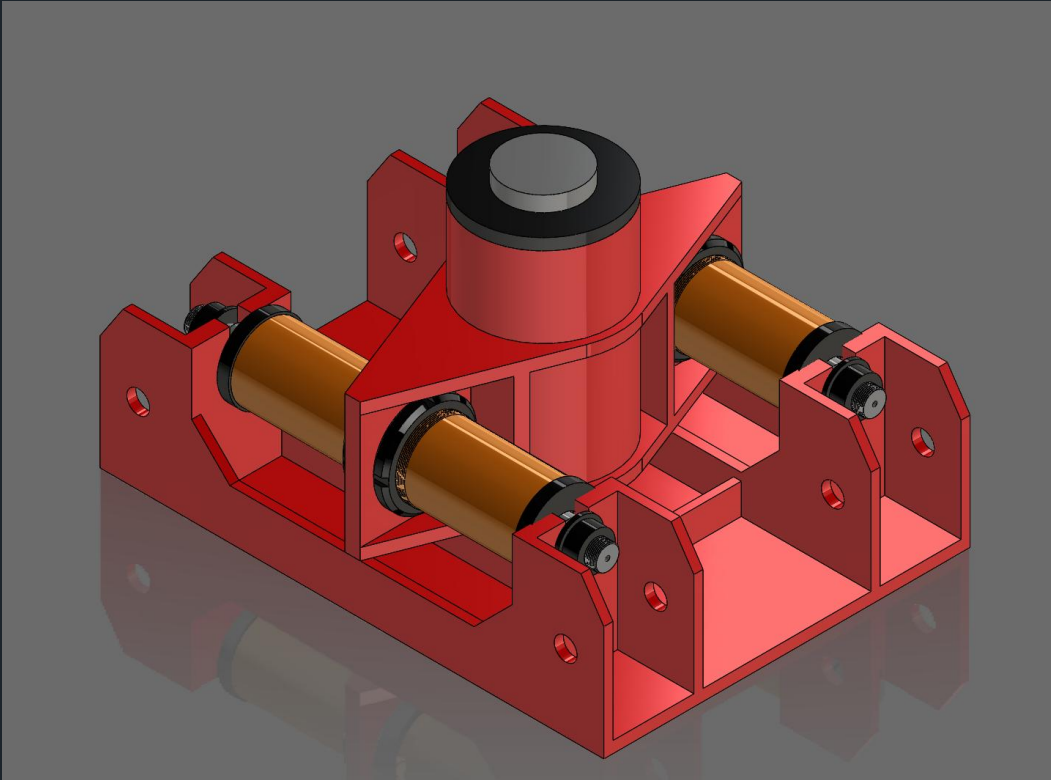
~1–2.5 t
TYPICAL FORK
CAPACITY

Parallel
LINKAGE GEOMETRY

Skid base
STABILITY FEATURE

Reference ranges based on common hydraulic pallet-fork and tractor-attachment lift classes.





HYDRAULIC POSITIONING

Horizontal Moving Jack

A compact housing carrying three parallel hydraulic cylinders on a common bracket — one primary ram plus two secondary rams. This arrangement is characteristic of jacking/skidding units used to translate heavy machinery, structural sections, or vessels horizontally in short, controlled, synchronized strokes.

KEY DESIGN FEATURES

- Three hydraulic cylinders sharing one bolted housing
- Twin lifting-ear brackets for crane handling of the whole unit
- Compact footprint suited to confined installation locations
- Synchronized ram strokes for controlled incremental travel

3 rams
CYLINDER
CONFIGURATION

Multi-ton
SKID/JACK CLASS

Sync.
ACTUATION
STRATEGY

General reference context for multi-cylinder hydraulic skidding/jacking units used in heavy-equipment relocation.

METAL FORMING

Pipe Bending Machine

A pedestal-mounted hydraulic pipe bender: an open steel stand supports a rotating bend head that carries the die and clamp block, actuated by two hydraulic cylinders. The pipe is clamped against the die, and the rotating arm sweeps the free end around it to form the bend to the required angle.

KEY DESIGN FEATURES

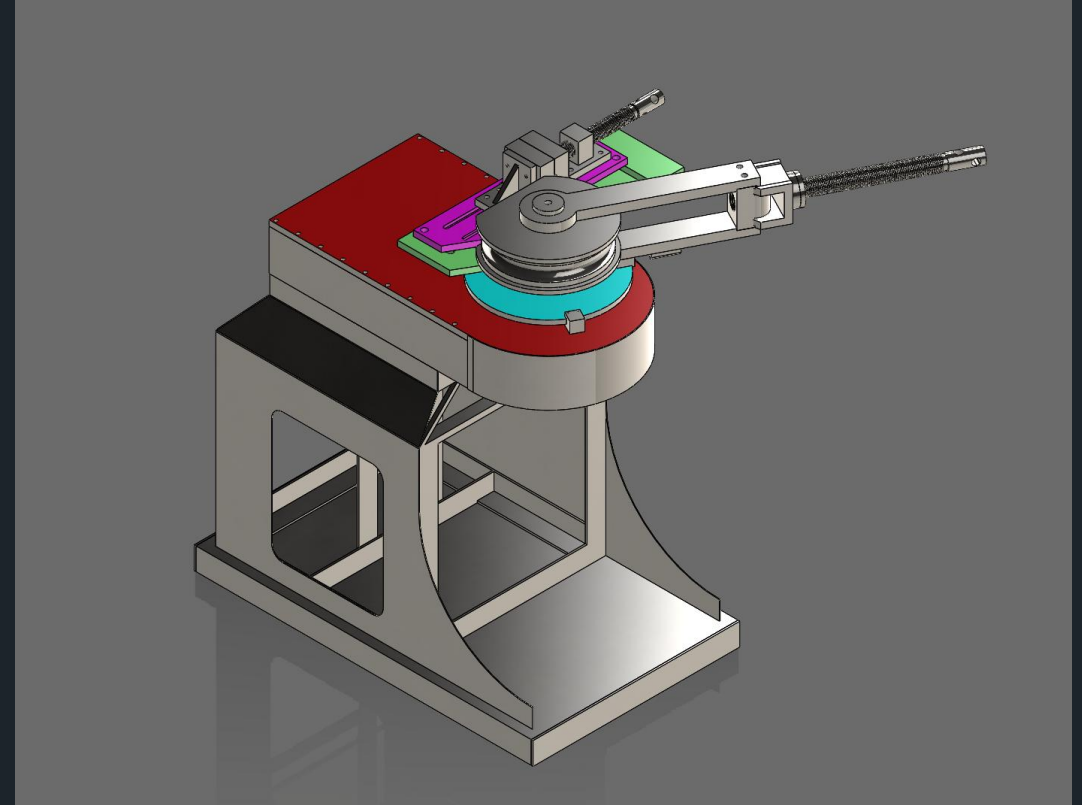
- Rotary bend-arm actuated by dual hydraulic cylinders
- Interchangeable bending die/mandrel for different pipe sizes
- Open pedestal base keeps swarf and long stock clear of the mechanism
- Sliding clamp block secures the pipe before the bend stroke begins

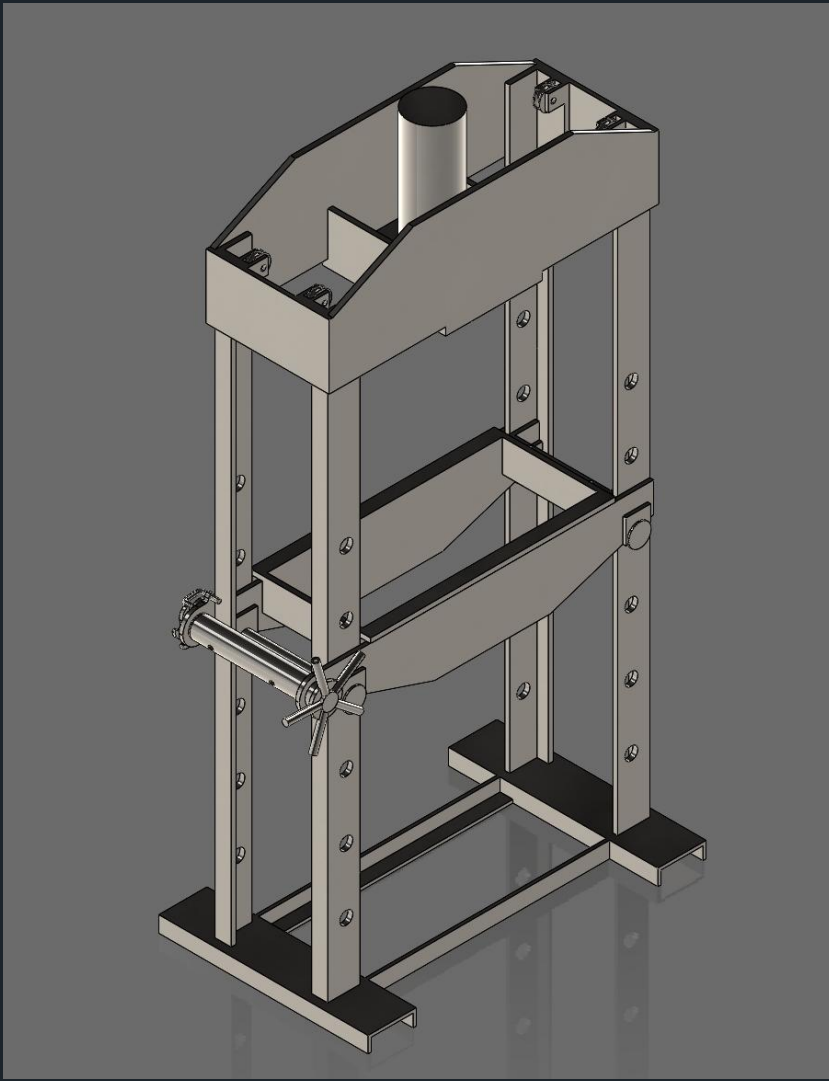
6–100 t
TYPICAL HYDRAULIC
CLASS

1/2–6 in
PIPE OD RANGE (BY
CLASS)

Up to 180°
ACHIEVABLE BEND
ANGLE

Reference ranges compiled from commercial hydraulic pipe/tube bender specifications (Harbor Freight, Hines Bending, Yash Machine Tools).





HYDRAULIC PRESS

Punch Press (H-Frame)

A classic four-column H-frame hydraulic press. The top beam carries the ram cylinder; the lower crossbeam — which holds the work bed — bolts to the columns at any of several height positions, so the working gap can be set for the job. A removable hand-pump jack unit provides the hydraulic force.

KEY DESIGN FEATURES

- Bolted 4-column H-frame for high stiffness under load
- Height-adjustable lower crossbeam / work bed
- Removable hand-pump hydraulic jack for portability and service
- Fully open sides for feeding long or wide workpieces

10–100 t
TYPICAL CAPACITY
CLASS

**150–200
mm**
TYPICAL RAM STROKE

**Multi-
position**
BED HEIGHT
ADJUSTMENT

Reference ranges based on common H-frame shop press classes (JEGS, Palmgren, VEVOR).

Sheet Metal Punch Press

The same H-frame press format, fitted here with a matched punch-and-die tooling set for blanking or piercing sheet metal. Guide pins on the die shoe keep the punch aligned to the die opening on every stroke, and the two-piece bolster (upper platen + lower die shoe) spreads the punching load evenly into the frame columns.

KEY DESIGN FEATURES

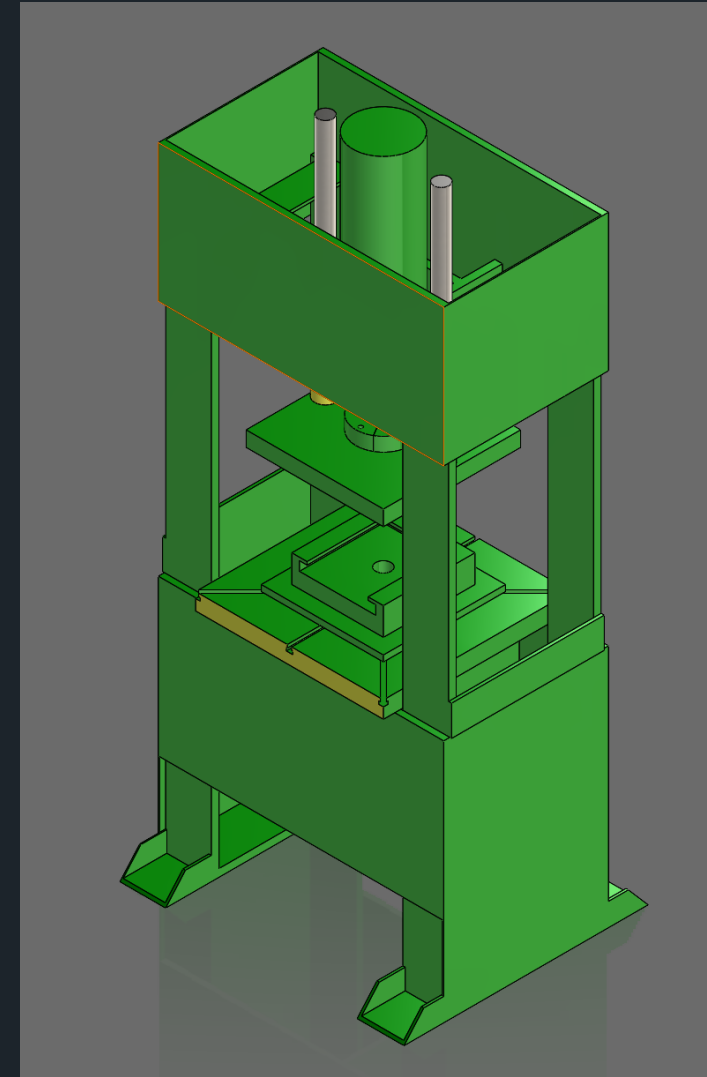
- Matched punch/die tooling set with guide-pin alignment
- Two-piece bolster plate (platen + die shoe) for even load transfer
- Guided ram rails prevent punch side-load and tooling wear
- Frame shared with the general-purpose press for tooling flexibility

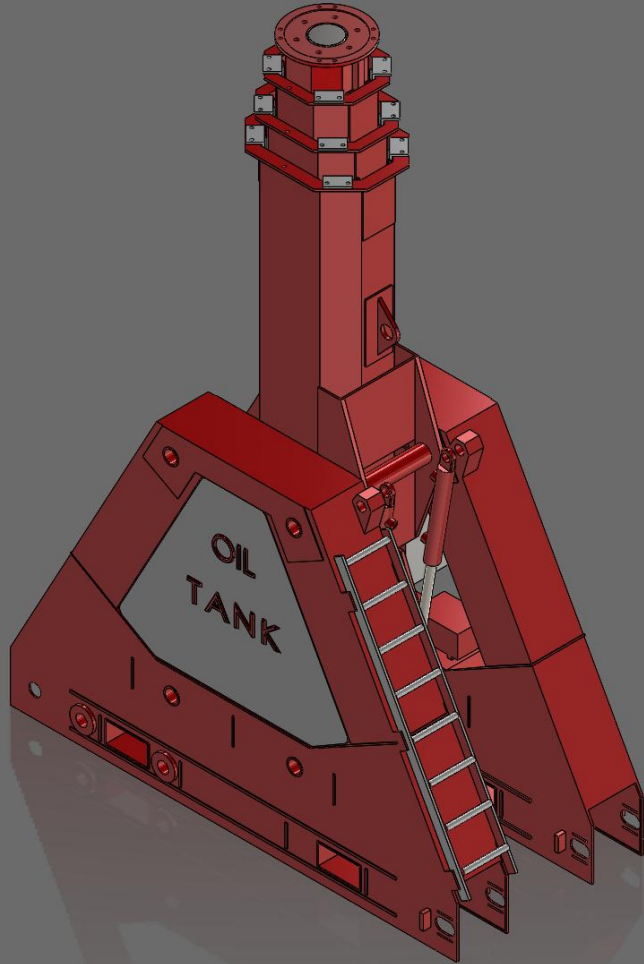
20–150 t
TYPICAL CLASS FOR
BLANKING

**Up to ~6–
10 mm**
MILD STEEL
THICKNESS

Guided
PUNCH/DIE
ALIGNMENT

Reference ranges for hydraulic punch/blanking press classes on H-frame architectures.





MOBILE PLATFORM SUBSYSTEM

Boom-Lift Oil Tank & Telescoping Mast

A structural sub-assembly in which the triangular base doubles as the hydraulic oil reservoir ('OIL TANK') for the machine, with a telescoping multi-stage mast rising from it. A bolted flange with a large bolt circle at the top mounts the boom or turret assembly, a small auxiliary cylinder handles locking/leveling, and a side ladder provides maintenance access to the upper stages.

KEY DESIGN FEATURES

- Triangular base structure integrated with the hydraulic oil reservoir
- Multi-stage telescoping mast for compact stowed height
- Bolted top flange with full bolt circle for the boom/turret interface
- Auxiliary hydraulic cylinder for locking or leveling functions
- Side-mounted maintenance ladder to the upper mast stages

Integrated
RESERVOIR + BASE

Multi-stage
TELESCOPING MAST

Bolt-circle
TOP MOUNTING
FLANGE

Functional description based on the modeled geometry; typical of mobile elevating work platform (MEWP) mast/reservoir subassemblies.

DESIGN SUMMARY

Common Threads Across the Portfolio

- Every mechanism uses hydraulic cylinders as the primary actuator — from single rams (jacks, presses) to multi-cylinder blocks (skidding unit, bending machine).
- Structural frames are simple, bolted, and serviceable: H-frames, mast/rail guides, and pinned linkages recur across scales, from portable jacks to press tooling.
- Alignment and guiding features (mast rails, guide pins, parallelogram linkages, clamp blocks) consistently protect the hydraulic ram from side-loading.